

Blocky Project Living Brief

Snapshot date: 2026-06-09

What This Project Is

Blocky is a build-to-learn blockchain engineering lab. You are learning by shipping real software, not only reading theory.

The project combines two tracks:

- Blockchain engineering track: protocol simulation, Solidity contracts, DeFi/NFT/marketplace modules, and multi-chain deployment.
- Multi-learning software track: learner progress data model, analytics views, and BI dashboards.

Original Goal

Build a professional portfolio-grade system where you can explain blockchain internals, write safer smart contracts with tests, build product-facing learning flows, and measure outcomes with data and dashboards.

Current Status (As Of 2026-06-09)

Repository and Workflow

- Repository structure is created.
- Docs-only execution model is active: you run commands and tools; Copilot maintains documentation and guidance.
- Baseline CI workflow file exists at `.github/workflows/ci.yml`.

Data and Analytics Progress

- Local SQLite database exists at `data/sqlite/learning.db`.
- Core schema exists: users, modules, lessons, checkpoints, submissions, events.
- Seed data confirmed: users 1, modules 2, lessons 4, checkpoints 8, submissions 0, events 0.
- BI views created: `v_completion_funnel`, `v_error_hotspots`.

Tableau Progress

- Tableau is connected to learning.db through ODBC (SQLite3 Datasource).
- v_completion_funnel has been dragged into the data model and rows are visible.
- Next charting work is to build sheet visuals from the two views.

Tools In Use

Core Dev

- Node.js and npm
- Git
- Python (via py launcher)
- PowerShell

Blockchain

- Foundry stack target: forge, cast, anvil

Data and BI

- SQLite database file at data/sqlite/learning.db
- DB Browser for SQLite
- Tableau Desktop via ODBC SQLite3 driver

What To Do Next

1. Tableau chart 1 (completion by module): source v_completion_funnel, X Module Name, Y Lesson Count or Submission Count, Color Topic.
2. Tableau chart 2 (fail hotspots): source v_error_hotspots, X check_type, Y fail_count.
3. Insert sample rows into submissions, then re-query both views and validate hotspot signal.
4. Capture screenshots and top 3 insights in runbooks.

Working Agreement

This is the living master brief. Update the markdown source each session and export a new PDF snapshot after meaningful updates.

Session Checkpoint Log

2026-06-09

- SQLite schema and seed checkpoint confirmed.
- ODBC connection to Tableau completed.
- v_completion_funnel loaded in Tableau data model.
- Next action: complete two Tableau visuals and capture insights.